Suggested Teaching Guidelines for

C++ Programming

PGCP-AC, August 2026

Duration: 90 Hours: 30 Theory (T) + 30 Lab (L) + 30 Self Learning (SL)

Objective: To learn Object Oriented Programming using C++.

Prerequisites: Knowledge of Computer Fundamentals.

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

List of Course Material

Text book:
Object-Oriented Programming in C++, 2008 edition, Rajesh K Shukla / Wiley

Reference book:
1. The C++ Programming Language, 4th edition, Bjarne Stroustrup / O'Reilly
2. Thinking in C++, 2nd edition, Bruce Eckel / Pearson

Note: *This module adopts C++ 17 as the programming standard for all sessions, lab work, and evaluations.*

Session 1: Getting Started (2T + 2SL)

- Installation and Setup of Development environment
- History of C++
- C++ versus C
- Features of C++
- C++ Program structure
- Introduction of advanced C++ concepts
- New features of C++ 17

Sessions 2: Beginning with C++ (2T + 4L)

- C++ Tokens
- Initialization
- Expressions
- Operators
  - Arithmetic Operator
  - Relational Operator
  - Logical Operator
  - Unary Operator
  - Ternary Operator
  - Assignment Operator

Lab:
- Setup the Development environment

- Write different C++ programs to:
  - Print Hello World
  - Add two numbers/binary numbers/characters
  - Calculate compound interest
  - Calculate power of a number
  - Swap two numbers
  - Calculate area of rectangle

Session 3: Conditional and Looping Statements (2T + 2L + 4SL)

- Conditional Statements: If, else if, switch
- Loops
  - for loop
  - while loop
  - do while loop
- Jump statement (break, continue & return keyword)
- Arrays
- Declaration and initialization of an array
- 1-D and 2-D arrays

Lab:
- Implement all control structures through your programs
- Implement a program which accepts command line arguments from main function.

Session 4: Functions (2T + 2L + 2SL)

- Different forms of functions
- Function prototyping
- Call by Reference
- Inline Functions
- Math library functions etc.

Lab:
- Implement functions through your program
- Declare function and call it by reference and note the observations
- Implement Inline functions in your program

Sessions 5 & 6: Memory Management and Pointers (4T + 4L + 4SL)

- Introduction to memory management in C++
- Pointers in C++
- Arrays using pointers
- Enumeration
- Typedef & Type Aliases

Suggested Teaching Guidelines for

C++ Programming

PGCP-AC, August 2026

- Using New operator
- Pointers to Objects: Class pointer
- this pointer
- Smart Pointers: unique_ptr, weak_ptr, shared_ptr
- Memory deallocation using delete operator

Lab:

- Assignments using pointers, arrays of pointers
- Assignments on passing pointers in functions
- Using pointers, write your own functions for the following:
  - String comparison
  - String concatenate
  - String copy
  - String length

Sessions 7 & 8: OOP Concepts, Constructor and Destructor (4T + 4L + 2SL)

- Object Oriented Programming Concepts
- Classes and Objects
- Access Specifiers, Inheritance, Polymorphism
- Namespaces
- Constructors & Destructors
- Parameterized constructors
- Multiple constructors in class
- Dynamic initialization of objects
- Copy Constructors
- Move semantics: Lvalue, Rvalue, move constructor
- Static Members
- Constant Members

Lab:

- Write a student class and use it in your program. Store the data of 10 students and display the sorted data according to their roll numbers, dates of birth, and total marks.
- Implement constructor and destructors through your program.

Session 9: Inheritance (2T + 2L + 4SL)

- Types of inheritance
- Single inheritance
- Multiple inheritance
- Multilevel inheritance
- Hierarchical inheritance
- Hybrid inheritance, etc.
- Virtual base class
- Constructors in derived class

Suggested Teaching Guidelines for

C++ Programming

PGCP-AC, August 2026

Lab:

- Design a hierarchy of computer printers. Use multiple inheritance in your hierarchy.
- Also use friend functions and classes in your program.

Session 10: Polymorphism (2T + 2L + 2SL)

- Types of Polymorphism
- Overloading functions
- Overloading Operators
- Friend functions
- Constant functions

Lab:

- Write Date and Time classes that allow you to add, subtract, read and print simple dates in dd/mm/yyyy and time in hh:mm:ss formats. Use function overloading in your program.
- Assignments to overload =, ==, +, ++, --, <<, >> and [ ] operators.

Session 11: Virtual Functions and Abstract Class (2T + 2L+ 4SL)

- Run Time Polymorphism
- Virtual Functions and Pure virtual functions
- dynamic_cast, static_cast, const_cast, reinterpret_cast
- Interfaces
- Abstract class

Lab:

- Implement Abstract classes in your program
- Using virtual and pure virtual functions implement hierarchy of computer printers
- Implement diamond problem with real life example

Session 12: Exception Handling (2T + 2L + 2SL)

- Exception Handling Introduction
- Exception handling – throwing, catching, re-throwing an exception
- Specifying exceptions etc.

Lab:

- Implement exceptions and do proper management through your program
- Implement Custom exception class

Suggested Teaching Guidelines for

C++ Programming

PGCP-AC, August 2026

Session 13: File Handling in C++ (2T + 2L)

- Managing Console I/O Operations
- C++ streams
- C++ stream classes
- Unformatted I/O operations
- Formatted I/O operations
- Managing output with manipulators
- Definition of file
- File handling in C++
- Doing read, write operation in files

Lab:

- Assignments on files doing different operations

Sessions 14 & 15: STL and RTTI (4T + 4L+ 4SL)

- Introduction to Templates
- Function Templates
- Class Templates
- Introduction to C++ Standard Template Library
- Working with Stack, Vector, Queue, Map
- Introduction to RTTI (Run-Time Type Information)

Lab:

- Assignments on templates and STL Library